Amjad Althabteh

Amjadjamalshadi@gmail.com | 734-657-3714 | LinkedIn | Portfolio | Github | Echo of Sacrifice | Grand Forge Games

SUMMARY

Performance-focused C++ systems engineer specializing in optimization, simulation systems, autonomous systems, graphics/rendering pipelines, and low-latency architecture. Experienced in building production-grade tools from scratch with a focus on GPU throughput, memory layout, cache efficiency, and profiling-driven performance.

EDUCATION

Wayne State University — ***Detroit, MI | 2026***
B.S. in Computer Science — **GPA: 3.76**

EXPERIENCE

Immersalab Graphics | *Detroit, MI* — ***09/25 – 02/26***
Graphics Engineer
- Reduced GPU idle time by ~30% using RenderDoc profiling and rendering pipeline optimizations.
- Designed internal C++/JSON tooling and asset pipeline; profiled and optimized GLSL shader compilation paths.
- Improved frame submission throughput and GPU resource utilization across rendering workloads.

UE Game Programmer C++| Yorktown, VA — **04/26 – 06/26**
Grand Forge Games
- Built a gameplay ability system in C++ using Unreal Engine GAS, supporting ability types with Blueprint integration.
- Developed AI behaviors and animation state machines for 4+ character types.
- Reduced animation transition overhead using multi-layer blending and reusable C++ systems.

Rare Munchies | *Ann Arbor, MI* — ***09/23 – 04/24***
Software Development Intern
- Reduced unnecessary re-renders by ~35% through improved state management and component optimization.
- Built reusable TypeScript component systems focused on runtime efficiency and minimal overhead.

Wayne State University | *Detroit, MI* — 0***2/25 – 12/25***
Undergraduate Researcher — Orbital Collision Modeling
- Simulated 50+ satellite trajectories using N-body gravitational dynamics and numerical integration across 1M+ evaluations.
- Accelerated simulation runtime by ~30% using spatial partitioning, scaling to large orbital collision datasets.

PROJECTS

Glint 3D Rendering Engine | *C++17/OpenGL* — ***Glint-Rendering***
- Designed a memory pool allocator using placement new and free list reuse, reducing heap allocation overhead by ~60%.
- Improved rendering performance using frustum culling and GPU instanced rendering, cutting draw calls by ~40%.
- Built custom rendering subsystems including shader, mesh, and texture management.

3D Space Simulation Engine — ***OrbitalCollision***
- Engineered a modular C++17/OpenGL simulation engine simulating 50+ orbital bodies across 1M+ trajectories.
- Reduced simulation runtime by ~40% using spatial partitioning and numerical integration.
- Developed a custom 3D math library for vector and matrix transformations with no external dependencies.

HFT OrderBook Engine | *C++20* | *Performance System* — ***HFT***
- Built a C++20 order matching engine with price-time priority achieving microsecond-level order resolution.
- Eliminated false sharing using cache-aligned (alignas(64)) structures, reducing memory contention under concurrent load.

Traceforge — Diagnostics Engine | *Stack Trace & Memory Tracking* — ***DiagnosticsEngine***
- Engineered a C++23 diagnostics engine using std::stacktrace for runtime debugging and crash analysis.
- Implemented custom allocation tracking using overloaded new/delete operators for memory leak detection.

SKILLS

Languages: C++ (C++11–C++23), Rust, C, C#, Python, TypeScript, Java, Javascript, SQL
Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot, vulkan
Systems: Linux , custom allocators, memory pools, multithreading, lock-free data structures, cache optimization, profiling